

Structure-Based Prediction of NES Motifs using Graph Neural Networks

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Abstract

This project addresses the prediction of Nuclear Export Signals (NES) in protein structures using a graph-based representation of protein 3D data. We convert PDB files into residue-level graphs and train Graph Neural Networks (GNNs) to classify whether a protein contains NES motifs. Each residue is represented as a graph node with features derived from the original 3D protein structure. Node features are encoded either as a one-hot representation of the amino acid type or using ESM-2 embeddings that capture contextual information from the protein sequence. Edges reflect spatial proximity, and the resulting data is suitable for modern geometric deep learning. Our pipeline enables direct use of structural information for biologically relevant predictions.

1 Introduction

Nuclear Export Signals (NES) are short, leucine-rich motifs that direct proteins from the nucleus to the cytoplasm via CRM1-mediated transport. Accurate NES prediction is crucial for understanding protein localization and function, particularly in viral and cancer biology. While most prior methods rely on sequence-based models, they often fail to capture structural constraints that influence motif accessibility and functionality.

In this work, we propose a structure-based approach leveraging Graph Neural Networks (GNNs) to classify NES-containing proteins using spatial graphs derived from 3D structures predicted by AlphaFold2, where Nodes represent amino acid residues, with features derived either from one-hot encodings or ESM-2 embeddings, and edges reflect spatial proximity. This formulation allows learning from both sequence and geometry, offering a richer context for NES detection. [1] [2] [3] [4]

2 Methods

2.1 Data & Pre-processing

We use protein structures predicted by AlphaFold2, provided in .pdb format. The dataset includes 675 proteins, each annotated as either NES-positive or NES-negative.

Each PDB file is parsed using BioPython to extract C-alpha atoms representing amino acid residues. Labels are assigned based on external CSV metadata mapping each protein to its NES status.

2.2 Graph Construction Pipeline

We developed a complete pipeline to convert a protein’s 3D structure (in PDB format) into a graph-based representation suitable for geometric deep learning with PyTorch Geometric. The primary objective was to focus on nuclear export signal (NES) regions — specifically residues from chain B, which represent the NES motif — and their surrounding structural environment. To do this, we first filter the protein structure to

retain only residues from chain B and any residues from other chains that lie within a 15 Ångström radius of it, based on the position of their C-alpha atoms. This filtered structure can be saved as a new PDB file for inspection in PyMOL. We then build a graph where each node corresponds to a residue (specifically its CA atom), and each node is encoded using a 20-length one-hot vector for its amino acid type, with an additional binary feature indicating whether the residue belongs to the NES chain (chain B) or not. Edges are created between pairs of residues if their C-alpha atoms are closer than 8 Ångström, and the edge attributes store the actual Euclidean distances. We implemented a visualization function that plots the protein graph in 2D, highlighting NES residues in red and nearby residues in blue, along with the spatial proximity edges. This entire preprocessing framework enables us to isolate biologically meaningful substructures and represent them as spatial graphs for downstream tasks like NES classification.

2.3 Model Selection Strategy

To identify the most suitable Graph Neural Network (GNN) architecture for NES classification, we began by comparing leading geometric models with respect to key criteria: geometric compatibility, data efficiency. Our evaluation centered on two candidates: GVP-GNN and EGNN.

Feature	GVP-GNN	EGNN
Core Representation	Scalar/vector features	Embeddings + coordinates
Equivariance	SE(3)-equivariant	Invariant to rotation, translation, reflection
Complexity	High (vector operations)	Moderate
Use of Geometry	Excellent for directional vectors	Excellent for distance-based edges
Data Efficiency	Requires large datasets	Performs well on small datasets
Scalability	Limited	Highly scalable

Table 1: Comparison of GVP-GNN and EGNN

Given that our input graphs are built from C-alpha atoms and include only distance-based edges—without directional vectors—EGNN provides a natural fit. It maintains key geometric invariances while requiring fewer data samples, making it well-suited for our dataset size and hackathon constraints. As such, we selected EGNN as our primary architecture. [6] [5] [7] [8]

2.4 Establishing a Baseline: Graph Convolutional Network (GCN)

Before deploying geometry-aware models, we implemented a basic Graph Convolutional Network (GCN) to validate our data pipeline and establish a performance baseline.

In GCN, each node aggregates information from its neighbors by computing a weighted average of their features. In our setting, each node corresponds to a residue (C-alpha atom), encoded via a one-hot amino acid vector, and edges represent proximity under an 8 Å threshold.

Advantages of GCN:

- Lightweight and easy to implement.
- Fast training and suitable for pipeline validation.
- Effective use of local connectivity.

Limitations:

- Does not incorporate spatial coordinates or distances.
- Lacks geometric invariance.
- May miss subtle 3D motifs critical for NES identification.

Conclusion: While GCN offered a solid starting point for model development, its lack of geometric sensitivity limited its effectiveness in this structural biology task. This motivated our shift to EGNN, which explicitly incorporates spatial structure into message passing.

3 Results and Model Evaluation

3.1 Protein Graph Visualization

We began by visualizing the full protein graphs, where each node corresponds to a residue and edges reflect spatial proximity. **Figure 1** illustrates an unfiltered graph, which includes all residues and edges, resulting in a dense and complex structure. To enhance interpretability and focus on NES-relevant biology, we filtered the graphs to retain only residues from chain B (typically containing the NES motif) and neighboring residues within a 15Å radius. The resulting focused graph, shown in **Figure 2**, is significantly sparser and more informative. Finally, in **Figure 3**, we represent the same NES-focused graph but with ESM-2 embeddings instead of one-hot features, allowing richer contextual encoding.

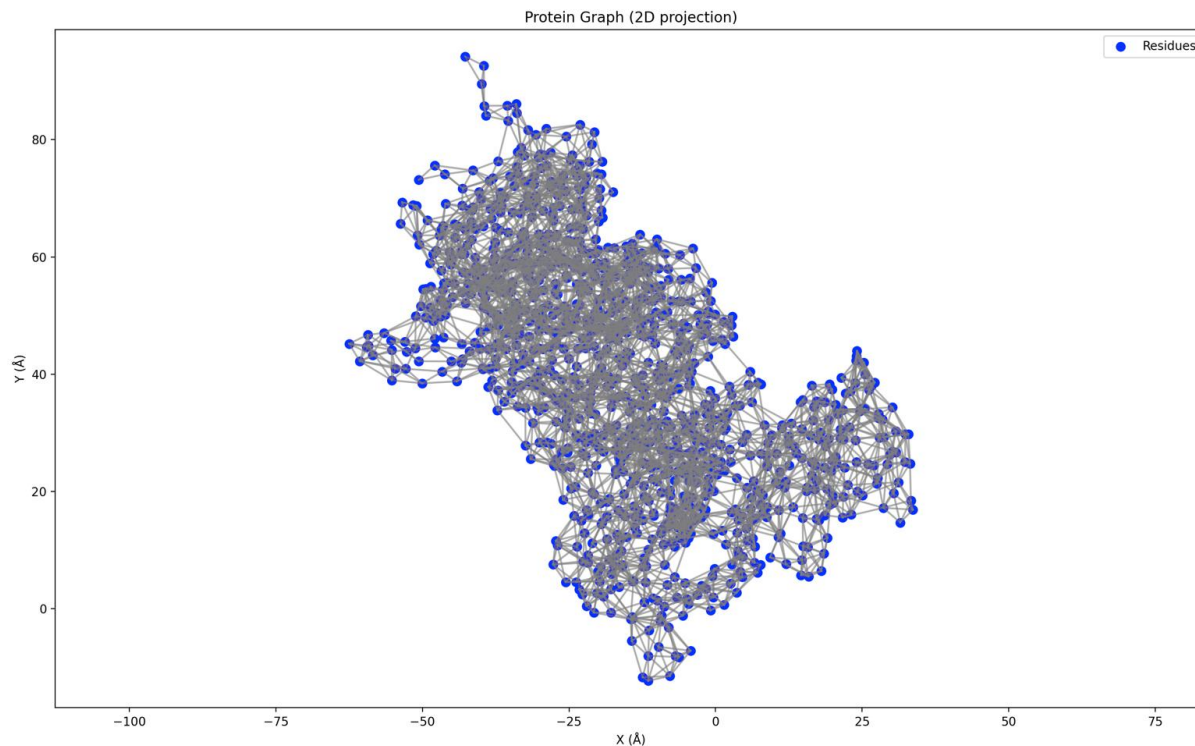


Figure 1: Initial dense protein graph including all residues.

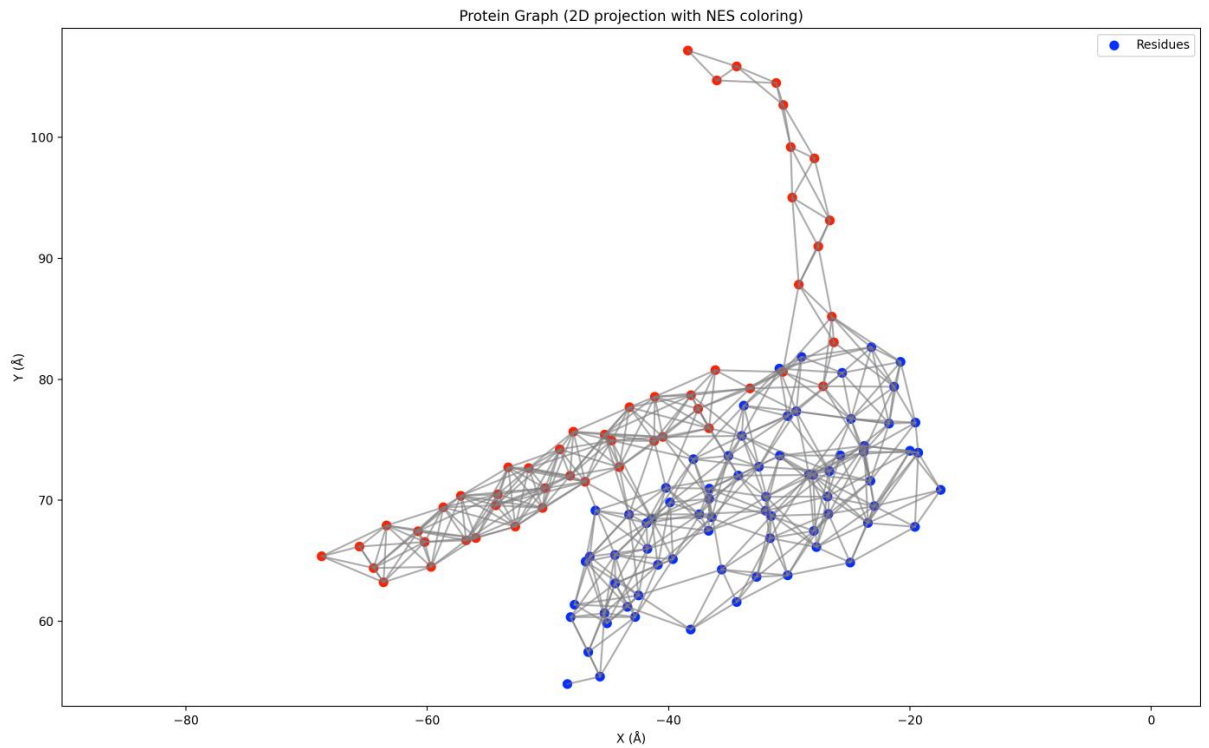


Figure 2: 2D projection of a NES-focused protein graph using one-hot encoded residues. Red: NES residues; Blue: proximal residues.

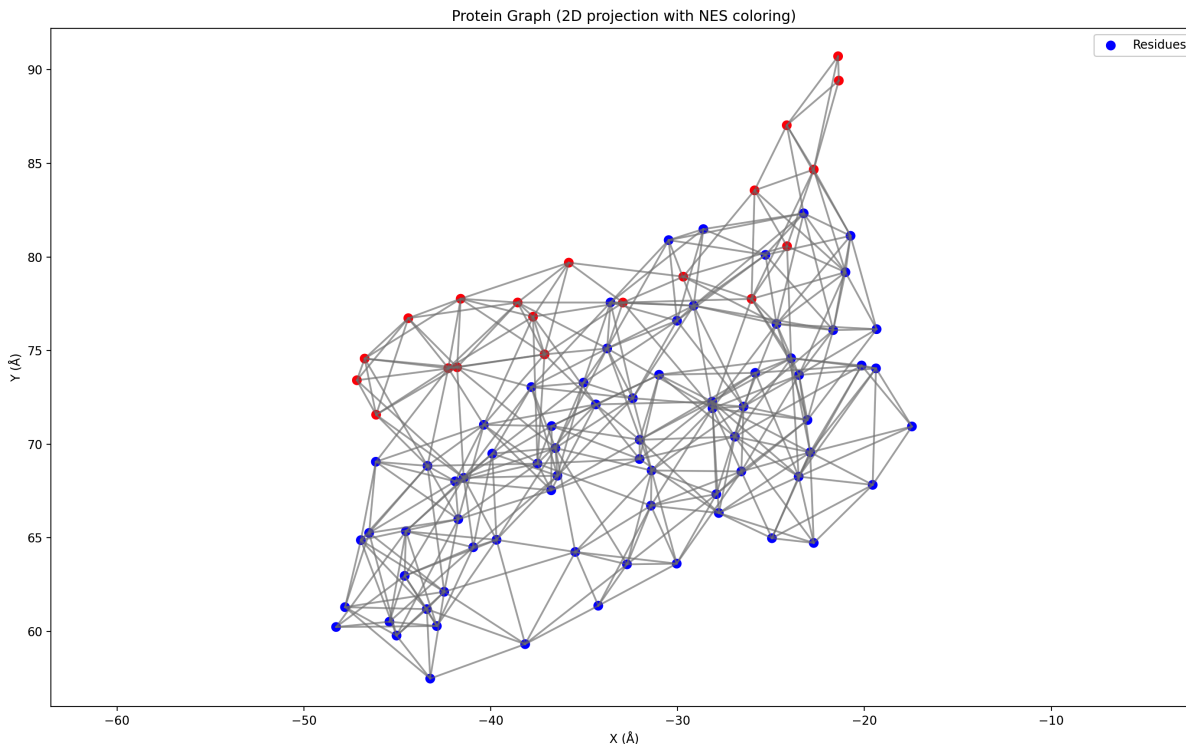


Figure 3: 2D projection of a protein graph showing residues embedded using esm2 from the NES chain (red) and nearby residues from other chains (blue), with edges representing spatial proximity.

3.2 Evaluation Metrics and Performance Visualization

To comprehensively assess model performance, we employed a suite of classification metrics. These include accuracy, precision, recall, F1 score, ROC-AUC, and additional tools such as the confusion matrix and prediction score distributions. Table 2 summarizes the purpose of each metric.

Metric	What It Measures	Why It Matters
Accuracy	Overall correctness	General performance
Precision	$TP / (TP + FP)$	Penalizes false positives
Recall	$TP / (TP + FN)$	Penalizes false negatives
F1 Score	Harmonic mean of precision and recall	Balances precision and recall
ROC AUC	Area under ROC curve	Discrimination capability
Confusion Matrix	Class-specific error distribution	Insight into misclassifications
PR Curve	Precision vs. recall trade-off	Better for imbalanced classes
Boxplot	Prediction score distribution	Model confidence and separation

Table 2: Evaluation metrics used and their relevance

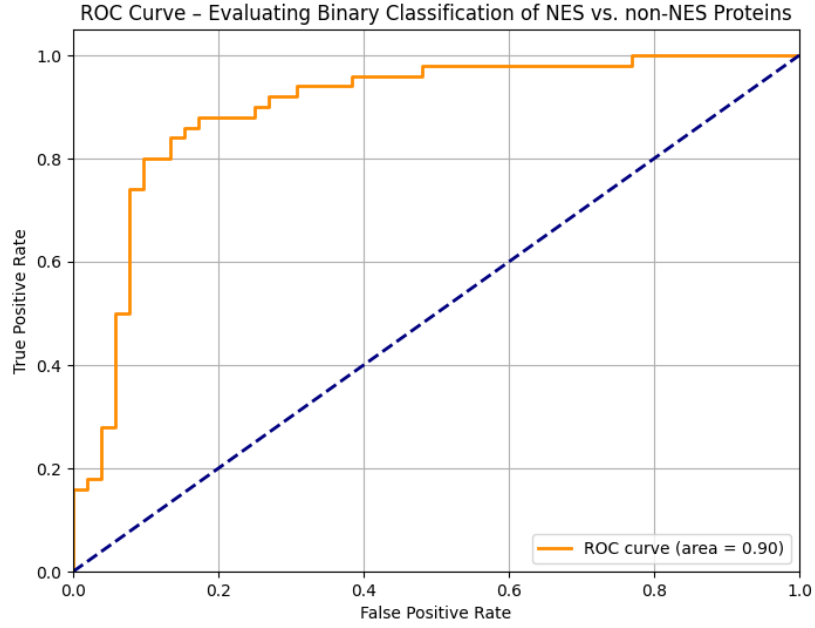


Figure 4: ROC curve: AUC = 0.90 for NES classification using EGNN.

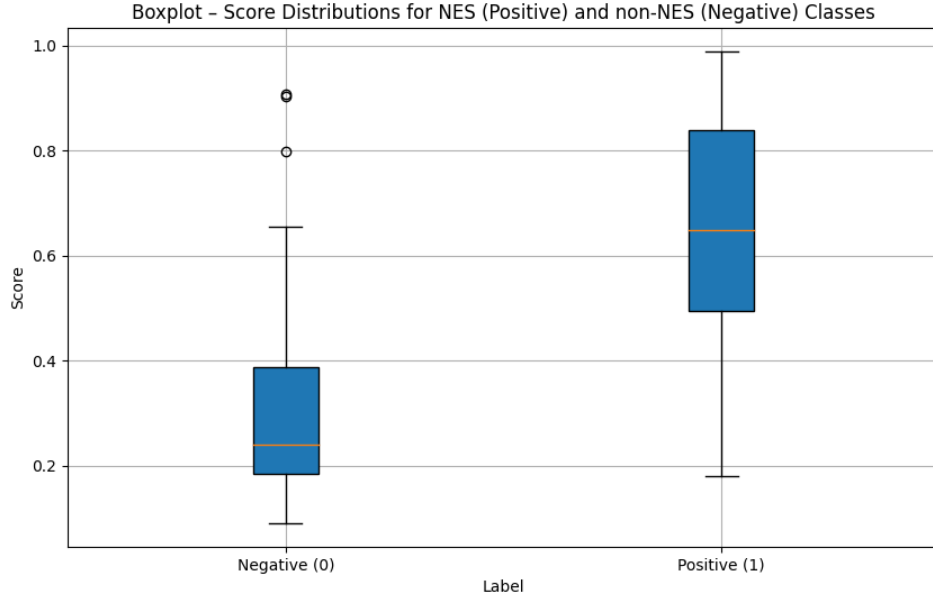


Figure 5: Boxplot of prediction scores: NES-positive proteins receive higher scores.

3.3 Model Diagnosis and Optimization Strategy

Comparison to Baseline. Our initial EGNN model was benchmarked against a high-performing reference model (AUC = 0.9148). While the reference model showed rapid convergence (loss ≈ 0.01 by epoch 50), our

model exhibited slower convergence with early signs of overfitting at epoch 23. Final performance: AUC = 0.90, accuracy = 84.2

Dropout Regularization. To address overfitting, we applied dropout at two levels: 0.3 and 0.5. Performance stabilized at $\text{AUC} \approx 0.90$, but the gains were modest, suggesting dropout alone was insufficient for substantial regularization.

Optimizer Exploration. We compared optimizers while maintaining fixed architecture:

- **Adam:** AUC = 0.90, Accuracy = 0.824.
- **AdamW:** Comparable to Adam, slightly more stable.
- **SGD with Momentum:** AUC = 0.65, Accuracy = 0.51. Slower convergence and unstable.

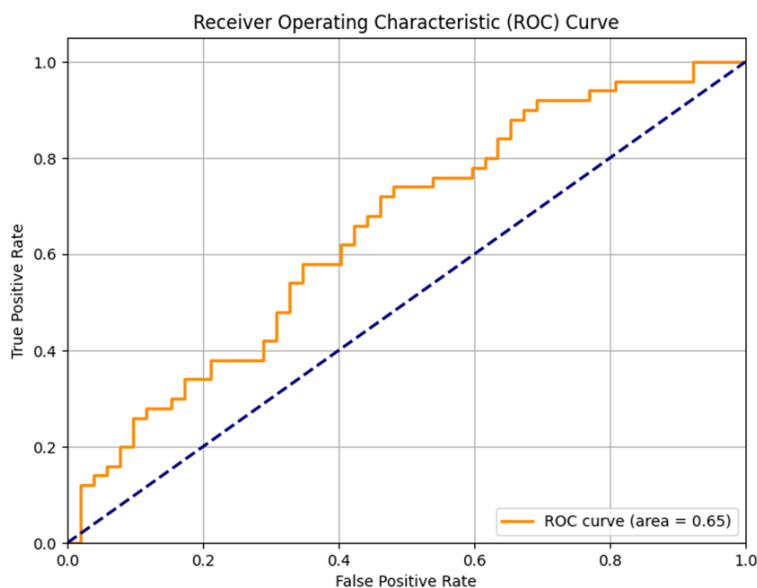


Figure 6: ROC for SGD with momentum. Shows poorer separability.

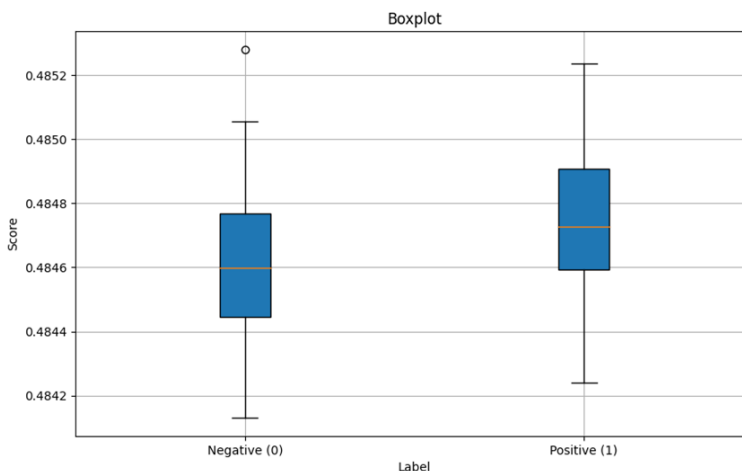


Figure 7: Boxplot for SGD: Poor separation of NES vs. non-NES.

ESM-based Embedding Pipeline. We implemented a modular framework to extract residue-level embeddings from ESM-2, supporting variable embedding sizes (e.g., 1280, 2560) and selectable layers (e.g., layer 30). All graphs were serialized into a single cached PyTorch file, streamlining experiment management.

Summary of Optimization Workflow.

- Identified overfitting via benchmark comparison.
- Applied dropout regularization.
- Conducted optimizer comparison.
- Built scalable ESM-embedding infrastructure.

Recommended Next Steps.

- **Augmentation:** Extend the dataset using AlphaFold-predicted structures in future work.
- **Regularization:** Tune dropout placement and magnitude; add weight decay.
- **Feature Representation:** Experiment with different ESM layers and combined embeddings.
- **Architecture:** Slightly increase hidden dimensions or add residual connections.
- **Training Dynamics:** Introduce learning rate schedulers and fine-tune optimizer parameters.
- **Evaluation:** Run comparative training regimes and visualize ROC and boxplot metrics.

4 Discussion

Our results highlight the potential of structure-based graph representations for predicting Nuclear Export Signals (NES) in proteins. By modeling proteins as spatial residue-level graphs and applying geometric deep learning, we successfully captured spatial patterns associated with NES functionality.

The EGNN model demonstrated strong performance ($AUC = 0.90$), confirming its suitability for processing protein structures derived from AlphaFold2 predictions. Its rotation and translation equivariant

architecture made it particularly effective for small-scale structural datasets. In contrast, the GCN baseline—although useful for validating our pipeline—lacked spatial inductive biases, leading to lower classification power.

Despite these achievements, overfitting was observed in later training epochs, as indicated by the divergence between training and validation loss. Attempts to mitigate this via dropout and optimizer tuning led to modest improvements, suggesting the need for more advanced regularization techniques or larger datasets. Furthermore, switching from one-hot encodings to pretrained ESM-2 embeddings introduced richer contextual information per residue, but added challenges in running time, memory allocations, tuning embedding dimensionality and selecting optimal internal layers.

Our findings underscore the importance of combining structural and sequence-derived information, and indicate that further exploration of embedding strategies, regularization, and attention mechanisms could lead to improved model generalization and interpretability. [10] [11]

5 Future Work

Building upon our current framework, we outline several directions for further development:

- **Embedding Optimization:** Experiment with layer aggregation strategies in ESM-2 (e.g., averaging or weighted sum of layers 26–33), or combine ESM with structure-confidence metrics such as pLDDT to enrich node features.
- **Advanced Regularization:** Introduce weight decay, dropout scheduling, or contrastive learning approaches to improve generalization and reduce overfitting.
- **Dataset Augmentation:** Extend the dataset by generating additional NES annotations from external databases or simulating NES-preserving mutations using structure-aware generative models.
- **Architectural Extensions:** Evaluate more expressive GNNs which may capture long-range dependencies and subtle geometric cues.
- **Generalization to Other Motifs:** Apply the pipeline to different signal motifs (e.g., NLS, phosphorylation sites) to assess adaptability and biological relevance across tasks.

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